CHAPTER 4

SAMPLING DISTRIBUTIONS

CHAPTER ESSENCE AT A GLANCE:
Master parameter versus statistic, sampling distribution concept, Standard Error (SE) formulas, Central Limit Theorem, and distribution shapes (t, F, Chi-Square).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Parameter vs Statistic:** Parameter is a fixed numerical characteristic of the entire population (mu, sigma, P); Statistic is a function of sample observations (x_{bar}, s, p) that varies across samples.

• **Sampling Distribution Concept:** The probability distribution of a statistic obtained from all possible random samples of a given size n drawn from a population.

• **Standard Error (SE):** The standard deviation of the sampling distribution of a statistic; measures sample precision and sampling variability (SE of sample mean = sigma divided by square root of n).

• **Central Limit Theorem (CLT):** As sample size n increases (n >= 30), the sampling distribution of sample mean x_{bar} approaches a Normal distribution regardless of the underlying population distribution shape.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{SE of Sample Mean} \iff \text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \quad [\text{or } \frac{s}{\sqrt{n}} \text{ when } \sigma \text{ unknown}]$$

Precision Index of Sample Mean Distribution

$$\text{SE of Sample Proportion} \iff \text{SE}(p) = \sqrt{\frac{P(1-P)}{n}} = \sqrt{\frac{PQ}{n}}$$

Standard Error of Sample Percentage and Proportion

$$\text{Central Limit Theorem} \iff \lim_{n\to\infty} Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

Asymptotic Normality of Sample Mean for Large Samples

$$\text{SE of Difference of Means} \iff \text{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Two-Sample Independent Difference Dispersion

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse Standard Deviation with Standard Error; Standard Deviation measures scatter of individual observations, while Standard Error measures scatter of sample statistics.

• **Alert:** Standard Error is inversely proportional to square root of sample size (SE propto 1 divided by sqrt(n)); to reduce SE by half, sample size must be quadrupled (4n).

• **Alert:** Finite Population Correction factor sqrt[(N - n) divided by (N - 1)] must be multiplied when sampling without replacement from a small finite population.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Parameter and Statistic with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Parameter:** A numerical descriptive measure calculated from the entire population; it is a fixed constant (e.g. Population Mean mu, Population Standard Deviation sigma). **[1 Mark]**

• **Statistic:** A numerical descriptive measure computed from sample observations; it is a random variable that varies from sample to sample (e.g. Sample Mean x_{bar}, Sample Standard Deviation s). **[1 Mark]**

2. What is meant by Standard Error of a statistic? State two important utilities of Standard Error. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The standard deviation of the sampling distribution of a statistic is known as its Standard Error (SE). **[1 Mark]**

• **Two Utilities:**
• Measures the precision and reliability of the sample estimate (smaller SE indicates greater reliability).
• Forms the foundation for constructing confidence intervals and conducting statistical hypothesis testing. **[2 Marks]**

3. State the Central Limit Theorem and explain its practical significance in statistics. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Statement:** If a random sample of size n is drawn from any population with mean μ and finite variance σ^2, the sampling distribution of sample mean x_{bar} approaches a normal distribution with mean μ and variance σ^2 divided by n as n becomes large ($n \geq 30$). **[1.5 Marks]**

• **Significance:** Allows investigators to use normal distribution techniques and Z-tests for large samples without requiring the parent population to be normally distributed. **[1.5 Marks]**

4. A random sample of 100 students is taken from a population with Standard Deviation 20. (a) Calculate the Standard Error of the sample mean. (b) What sample size is required to reduce the Standard Error to 1? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) SE Calculation:** SE = sigma divided by square root of n = 20 divided by square root of 100 = 20 divided by 10 = 2. **[1.5 Marks]**

• **(b) Sample Size for SE = 1:** 1 = 20 divided by square root of n. square root of n = 20, so $n = 20^2$ = 400 students. **[1.5 Marks]**

5. In a city, 40 percent of residents are vegetarian. If a random sample of 400 residents is surveyed, calculate the Standard Error of the sample proportion. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Parameters:** P = 0.40, Q = 1 - 0.40 = 0.60, n = 400. **[0.5 Mark]**

• **Formula:** SE(p) = square root of [PQ divided by n] = square root of [(0.40 * 0.60) divided by 400] = square root of [0.24 divided by 400] = square root of 0.0006 = 0.0245 (or 2.45 percent). **[1.5 Marks]**

6. Name three exact sampling distributions used when sample size is small and the population variance is unknown. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Three Exact Distributions:** (1) Student's t-distribution; (2) Snedecor's F-distribution; (3) Chi-Square distribution. **[2 Marks]**